

SQ1

1. Based on the figure 1, what measurement is the arrow showing on the ruler?

- a. 61 cm
- b. 61 mm
- c. 6.0 cm
- d. 60 mm

2. Convert the length measurement of 406cm to mm.

- a. 406 mm
- b. 4600 mm
- c. 4.06 mm
- d. 4060 mm

3. Based on the Table 1, what would be the area of Trial 3 in sq.cm?

- a. 64.005
- b. 62.45
- c. 62.37
- d. 62.6

4. Which physical quantity is calculated using the formula

$$\rho = m / V$$

- a. Mass
- b. Volume
- c. Density
- d. Height

5. Which instrument gives the most precise measurement for small diameters?

- a. Ruler
- b. Micrometer Caliper
- c. Vernier Caliper
- d. Meter Stick

6. The percentage error of a measurement is used to determine the:

- a. Accuracy of the measurement
- b. Precision of the instrument
- c. Size of the object
- d. Density only

SITUATION 1: Volume Calculation

The mean outside diameter of a hollow cylinder is 2.80 cm, the mean inside diameter is 2.20 cm, and the height is 10.0 cm.

7. What is the correct formula for calculating the volume of a hollow cylinder?

- a. $V = \pi r^2 h$
- b. $V = \pi(R - r)h$
- c. $V = \pi h(R^2 - r^2)$
- d. $V = 2\pi r h$

8. Using the given measurements, what is the volume of the hollow cylinder in cubic centimeters (cm³)?

- a. 22.0 cm³
- b. 24.1 cm³
- c. 26.4 cm³
- d. 28.7 cm³

9. Why does the micrometer caliper generally give a smaller percentage error compared to the ruler?

- a. Because the micrometer caliper is easier to use than the ruler.
- b. Because the micrometer caliper has finer scale divisions and higher precision.
- c. Because the micrometer caliper measures longer objects better.
- d. Because the ruler is not a measuring instrument.

SITUATION 2: Percentage Error

The theoretical density of plastic is 1.35 g/cm³, while the experimental density obtained using the Vernier caliper is 1.30 g/cm³.

10. What is the percentage error of the experimental measurement?

- a. 2.5%
- b. 3.7%
- c. 4.0%
- d. 5.5%

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11. What does this percentage error indicate about the measurement?

- a. The experimental value is completely inaccurate.
- b. The measurement is close to the theoretical value and fairly accurate.
- c. The Vernier caliper is not suitable for density measurement.
- d. The experimental value is higher than the theoretical value.

12. What does the SI system stand for?

- a. Standard International System
- b. Scientific Instrument System
- c. Système International d'Unités
- d. Standardized Instrument Units

13. The micrometer caliper is primarily used to measure objects:

- a. With very large dimensions
- b. Roughly and quickly
- c. With high accuracy that fit between its spindle and anvil
- d. Only their length

14. The formula shown in the image for the volume of a marble (sphere) is:

- a. $V = \pi r^2 h$
- b. $V = 4/3 \pi r^3$
- c. $V = \pi/6 D^3$
- d. $V = \pi D h$

15. Which is the best reason why multiple trials are conducted mainly to:

- a. Change instruments
- b. Reduce measurement error
- c. Increase mass
- d. Measure time

FIGURE 1

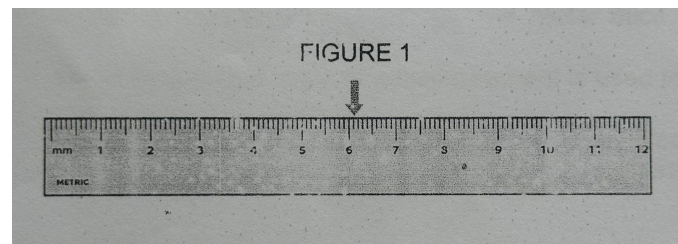


TABLE 1

Trial	Length (cm)	Width (cm)
1	12.55	5.10
2	12.49	5.00
3	12.60	4.95
4	12.52	5.08

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1	12.55	5.10
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